Chapter 3: Methodology

3.1 System Overview

This chapter presents the methodology for developing and evaluating a Deep Deterministic Policy Gradient (DDPG) agent for autonomous lane keeping, augmented with external driving dataset integration. The system operates in a dual-mode simulation environment: a high-fidelity CARLA 0.9.16 simulator with four-wheel PhysX vehicle dynamics for primary training and evaluation, and a nonlinear bicycle model for rapid prototyping and offline result generation. Both backends produce an identical eight-dimensional observation vector, enabling seamless policy transfer between simulation fidelity levels.

The complete pipeline comprises five stages:

1. **Dataset preloading** (Phase 0): Three external driving datasets are parsed, normalized, and injected into a stratified replay buffer.
2. **Training** (Phases 1–3): A 600-episode curriculum progressively introduces harder road scenarios whilst transitioning the buffer sampling weights from expert-dominant to simulation-dominant.
3. **Deterministic evaluation**: 100 episodes (20 per scenario × 5 scenarios) with noise disabled.
4. **Metrics computation**: ISO 15622:2018, IEEE 2846-2022, and UNECE WP.29 R157 compliance assessment.
5. **Publication figure generation**: Eight IEEE-format figures at 300 DPI.

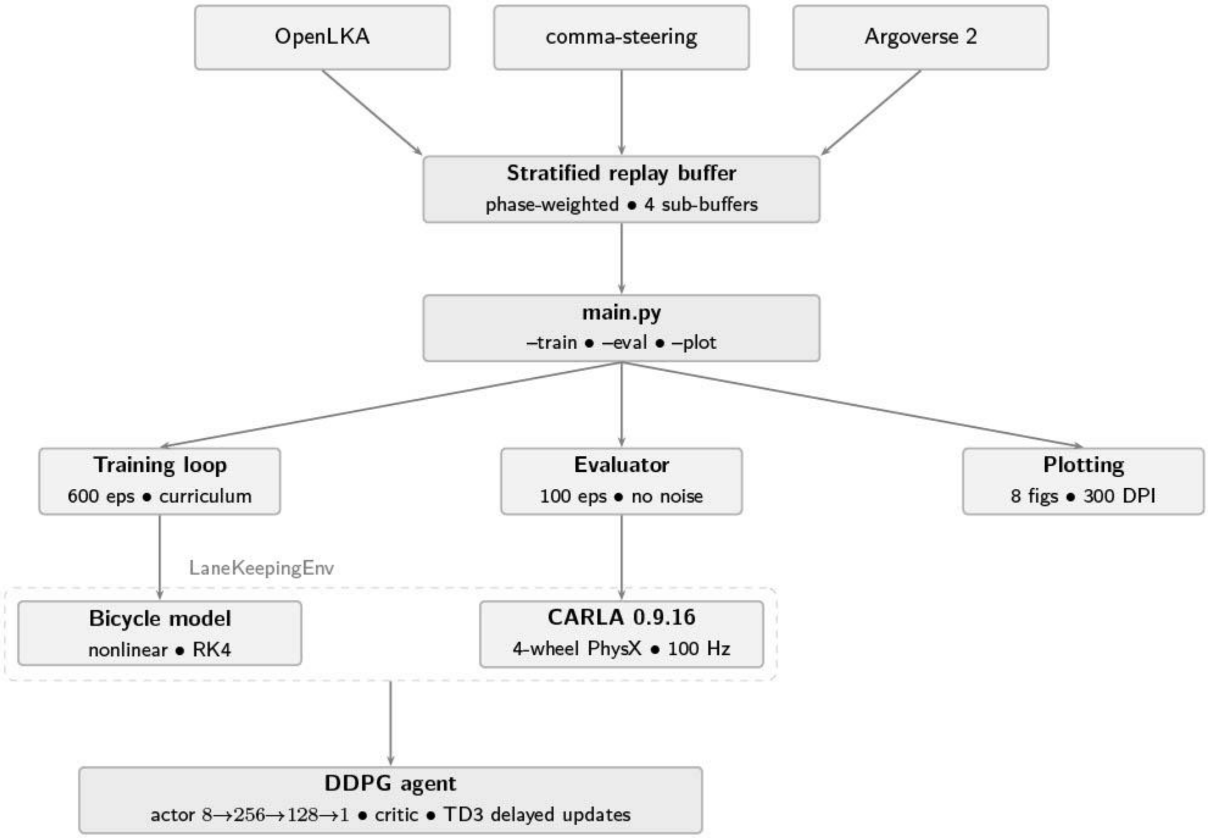

Fig. 3.1 System overview of the DDPG lane keeping pipeline

3.2 Simulation Environment

3.2.1 CARLA Simulator Configuration

The primary simulation environment employs CARLA version 0.9.16 (September 2025), an open-source autonomous driving simulator built on Unreal Engine 4.26 with PhysX-based vehicle dynamics. CARLA provides a physically accurate rendering pipeline and a comprehensive Python API for programmatic control of vehicles, sensors, and world state.

The simulator operates in synchronous mode with a fixed timestep of $\Delta t = 0.01 s (100 Hz)$, ensuring deterministic physics integration. This matches the control frequency specified in ISO

15622:2018 §8.1 for lane keeping assistance system evaluation. The synchronous mode guarantees that no physics steps are skipped between agent decisions, which is critical for reproducible reinforcement learning experiments.

Table 1: Simulation Parameters

Parameter	Value	Justification
CARLA version	0.9.16	Latest stable (September 2025)
Rendering engine	Unreal Engine 4.26	PhysX stability
Physics timestep	$\Delta t = 0.01s(100Hz)$	ISO 15622 §8.1 control rate
Synchronous mode	Enabled	Deterministic integration
Rendering mode	Enabled	Visual debugging capability
Traffic manager	Disabled	Ego-only evaluation

Table 2: Map-Scenario Mapping

Scenario	CARLA Map	Road Geometry
SCN-01	Town04	Long straight highway segments
SCN-02	Town03	Constant-radius curves (R = 80 m)
SCN-03	Town07	Winding rural roads
SCN-04	Town02	Narrow lanes for double lane change
SCN-05	Town01	Mixed urban with curves and straights

3.2.2 Four-Wheel Vehicle Dynamics

The ego vehicle employs the Tesla Model 3 blueprint (`vehicle.tesla.model3`) in CARLA, representing a Battery Electric Vehicle (BEV) midsize sedan. CARLA's PhysX engine simulates the vehicle with an independent four-wheel dynamics model that includes:

- Independent tyre forces for each wheel (front-left, front-right, rear-left, rear-right)
- Ackermann steering geometry with configurable maximum steering angle
- Suspension spring-damper model per wheel
- Aerodynamic drag and rolling resistance
- Drivetrain model with torque curve and automatic gearbox

The vehicle physics parameters are configured to match the representative BEV sedan class:

Vehicle Parameters Table

Parameter [1, 2, 3, 4]	**Symbol**	**Value**
Kerb mass	m	$1650\ kg$
Yaw moment of inertia	I_z	$2315.3 kg \cdot m^2$
CoM to front axle	L_f	1.105 m
CoM to rear axle	L_r	1.738 m
Wheelbase	L	2.843 m

Max front wheel angle	$\delta_{(max)}$	0.35 rad (20°)
Front cornering stiffness	$C_{(\alpha f)}$	88,000 N/rad
Rear cornering stiffness	$C_{(\alpha r)}$	94,000 N/rad
Reference speed	$V_{(ref)}$	16.67 m/s (60 km/h)

The tyre cornering stiffness values ($C_{\alpha f}, C_{\alpha r}$)are initially set to nominal) but may be overridden by the DS-02 calibration pipeline (§3.4.4) if the tyre model fit achieves $R^2 > 0.85$.

The four-wheel physics control is applied through CARLA's `VehiclePhysicsControl` API:

$$F_{(wheel,i)} = -C_{(\alpha,i)} \cdot \alpha_i \text{ for } i \in fl, fr, rl, rr$$

where α_i is the slip angle at each wheel, computed internally by PhysX from the vehicle state and tyre contact patch geometry.

3.2.3 Bicycle Model Fallback

For rapid prototyping and offline result generation without a running CARLA server, the system includes a nonlinear bicycle model with fourth-order Runge-Kutta (RK4) integration. This model is a two-degree-of-freedom approximation that captures the essential lateral dynamics of a four-wheel vehicle.

State vector ($x \in \mathrm{R}^8$):

$x = \left[X, Y, \psi, v_x, v_y, r, e_{(lat)}, e_\psi\right]^T$

where X, Y are global position, ψ is yaw angle, v_x is longitudinal velocity (held constant), v_y is lateral velocity, r is yaw rate, $e_{(lat)}$ is lateral deviation from lane Centre, and $e_{(\psi)}$ is heading error.

Equations of motion:

$$(\dot{v})_y = \left(F_{(yf)} + F_{(yr)}\right)/m - v_x \mathrm{r}$$

$$(\dot{r}) = \left(l_f \cdot F_{(yf)} - l_r \cdot F_{(yr)}\right)/I_\mathrm{z}$$

$$(\dot{e})_{(lat)} = v_x \sin\left(e_\psi\right) + v_y \cos\left(e_\psi\right)$$

$$(\dot{e})_\psi = r - \kappa_{(ref)} \cdot v_x$$

Tyre Model Force Equations

$$\alpha_f = \delta - \arctan\left(\left(v_y + l_f \cdot r\right)/v_x\right)$$

$$\alpha_r = -\arctan\left(\left(v_y - l_r \cdot r\right)/v_x\right)$$

$$F_{(yf)} = -C_{(\alpha f)} \cdot \alpha_f, F_{(yr)} = -C_{(\alpha r)} \cdot \alpha_r$$

Runge-Kutta 4 (RK4) Integration Steps

RK4 integration (mandatory Euler integration is insufficient for stiff tyre dynamics):

$$k_1 = f(x_t, u_t)$$

$$k_2 = f(x_t + (\Delta t)/2 \cdot k_1, u_t)$$

$$k_3 = f(x_t + (\Delta t)/2 \cdot k_2, u_t)$$

$$k_4 = f(x_t + \Delta t \cdot k_3, u_t)$$

$$x_{(t+1)} = x_t + (\Delta t)/6 \cdot (k_1 + 2k_2 + 2k_3 + k_4)$$

3.2.4 Road Scenario Design

Five road scenarios are defined for progressive difficulty evaluation,

Road Scenario Design Table

ID	Name	Geometry	Length
SCN-01	Straight Road	$\kappa = 0$	300 m
SCN-02	Constant Radius	$\kappa = 1/80m^{(-1)}$	500 m
SCN-03	Sinusoidal Winding	$\kappa(s) = 0.02\sin\left((2\pi s)/100\right)$	400 m
SCN-04	Double Lane Change	ISO 3888-2 exact geometry	175 m
SCN-05	Combined Urban	Multi-segment with S-bend	360 m

The SCN-04 Double Lane Change follows the exact geometry of ISO 3888-2:2011: a 50 m approach straight, a sinusoidal lateral displacement of \Delta y = 3.5 m over 25 m longitudinal distance, a 25 m corridor at offset, a return maneuver, and a 50 m exit straight. The lateral displacement profile is:

$$y(s) = (\Delta y)/2 \cdot \left(1 - \cos\left(\left(\pi \cdot (s - s_0)\right)/L_c\right)\right)$$

where L_c= 25 m is the lane change length.

3.3 DDPG Agent Architecture

3.3.1 Actor-Critic Networks

The actor-critic architecture operates on normalized observations and outputs continuous steering commands.

Actor network $\mu_\theta: \mathrm{R}^8 \rightarrow [-1,1]$:

Layer	Dimensions	Activation
Input	8	—
Hidden 1	256	ReLU + LayerNorm
Hidden 2	128	ReLU + LayerNorm
Output	1	tanh

Critic network Q_ϕ: $\mathrm{R}^8 \times \mathrm{R}^1 \rightarrow$:

Layer	Dimensions	Activation
Input (state)	8	—
Hidden 1	256 + 1 (action injected)	ReLU + LayerNorm
Hidden 2	128	ReLU + LayerNorm
Output	1	Linear

Weight initialization follows the fan-in scheme. Hidden layers are initialized uniformly in $\left[-1/\sqrt{(f)}, 1/\sqrt{(f)}\right]$ where f is the fan-in, and the output layer in $\left[-3 \times 10^{(-3)}, 3 \times 10^{(-3)}\right]$.

3.3.2 Delayed Policy Updates

Following TD3, the actor is updated only every d = 2 critic updates. This reduces variance in the policy gradient by ensuring the critic is more accurately estimated before the actor adapts.

Critic update (every step):

$$y = r + \gamma\left(1 - d_{(done)}\right) \cdot Q_{(\phi')}\left(s', \mu_{(\theta')}(s')\right)$$

$$L_{(critic)} = 1/N \sum (i = 1)^N \left(Q_\phi(s_i, a_i) - y_i\right)^2$$

Actor update (every d step):

$$\nabla_\theta J \approx 1/N \sum (i=1)^N \nabla_a Q_\phi(s,a)|_{(a=\mu_\theta(s))} \cdot \nabla_\theta \mu_\theta(s)$$

Target network update (Polyak averaging, every d steps):

$$\theta' \leftarrow \tau\theta + (1-\tau)\theta', \phi' \leftarrow \tau\phi + (1-\tau)\phi'$$

3.3.3 Ornstein-Uhlenbeck Noise with Annealing

Exploration employs an Ornstein-Uhlenbeck (OU) process to generate temporally correlated noise, which produces smoother steering exploration than uncorrelated Gaussian noise:

$$dx_t = \theta(\mu - x_t)dt + \sigma\sqrt{(dt)} \cdot N(0,1)$$

where $\theta = 0.15$ is the mean reversion rate and $\mu = 0$ is the long-run mean. The volatility σ is annealed linearly:

$$\sigma(e) = \{■(0.15\&e < 100@0.15 + (e-100)/350 \cdot (0.03 - 0.15)\&100 \leq e < 450@0.03\&e \geq 450))$$

3.4 Hybrid Data Fusion Pipeline

3.4.1 External Dataset Integration

Three publicly available driving datasets are integrated to seed the replay buffer with real-world driving transitions, reducing the sample complexity of pure simulation training:

ID	Dataset	Source	Content

DS-01	OpenLKA	UVA/NHTSA	Expert LKA demonstrations with CAN signals
DS-02	comma-steering-control	comma.ai	Steering angles, IMU, vehicle dynamics
DS-03	Argoverse 2	Argo AI	Trajectory data with HD map centerlines

Each dataset adapter parses raw sensor data into a canonical Raw Transition structure containing:

$$t = \left(e_{(lat)}, e_{\psi}, \kappa, v_y, r, \delta_{(prev)}, \kappa_{(la1)}, \kappa_{(la2)}, a, r_{(reward)}, s', d_{(done)}\right)$$

3.4.2 Universal Normalizer

All observations from both real datasets and simulation are normalized by physical limits rather than data-driven statistics. This guarantees zero distribution shift between sources:

$$\widehat{(o)}_i = o_i / c_i, \text{ clipped to } [-1,1]$$

Normalization Settings Table

Dimension	**Physical quantity**	**Normalization constant $(\mathbf{c_i})$**
0	$e_{(lat)}$	$L_w/2 = 1.75m$
1	$e_{(\psi)}$	$\pi/\ 4\ =\ 0.785\ rad$
2	κ	$0.05m^{(-1)}$
3	v_y	2.0 m/s
4	r	0.5 rad/s
5	$\delta_{(prev)}$	$\delta_{(max)} = 0.35rad$
6	$\kappa_{(la1)}$	$0.05m^{(-1)}$
7	$\kappa_{(la2)}$	$0.05m^{(-1)}$

A 5× physical limit check rejects transitions where any raw value exceeds five times its normalization constant, filtering parsing errors without discarding valid extreme maneuvers.

3.4.3 Stratified Replay Buffer with Phase-Aware Sampling

The Hybrid Stratified Buffer maintains four independent sub-buffers — one per data source — and samples from them with configurable, episode-dependent weights:

Phase Sampling Weights Table

Phase	Episodes	OpenLKA	Comma	Argoverse	Simulation
Phase 1	0–150	0.40	0.20	0.20	0.20
Phase 2	150–300	0.30	0.15	0.15	0.40
Phase 3	300–600	0.15	0.10	0.10	0.65

The phase transition ensures that:

1. Phase 1: The agent bootstraps from expert demonstrations, reducing initial random exploration.
2. Phase 2: Simulation data grows as the agent improves, balancing real and synthetic experience.
3. Phase 3: The policy is fine-tuned primarily on simulation rollouts matching the evaluation distribution.

Sub-buffers with zero entries are excluded from sampling and their weight is redistributed proportionally to non-empty sources. This allows the system to operate without any specific dataset by using the --skip-ds0X flags.

3.4.4 Tyre Model Calibration from Real-World Data

The DS-02 (comma-steering-control) adapter performs tyre model calibration by fitting effective cornering stiffness from real-world steering angle and lateral acceleration data:

$a_y \approx C_{(eff)}/m \cdot$

where $C_{(eff)}$ is the effective cornering stiffness. A least-squares fit yields:

$$C_{(eff)} = \left(\sum(i)\delta_i \cdot a_{(y,i)}\right) / \left(\sum(i)\delta_i^2\right) \cdot m$$

The front-rear split is estimated assuming the ratio $C_{(\alpha r)}/C_{(\alpha f)} \approx l_f/l_r$:

$$C_{(\alpha f)} = C_{(eff)} \cdot l_r/L, C_{(\alpha r)} = C_{(eff)} \cdot l_f/L$$

If the fit achieves $R^2 > 0.85$, the calibrated values replace the nominal tyre parameters in both the bicycle model and CARLA's WheelPhysicsControl.

3.5 Reward Function Design

The reward function comprises five components, each addressing a distinct control objective:

$$r_t = w_{(lat)} \cdot r_{(lat)} + w_{(head)} \cdot r_{(head)} + w_{(smooth)} \cdot r_{(smooth)} + w_{(prog)} \cdot r_{(prog)} + r_{(term)}$$

Reward Function Components Table

Component	Weight	Formulation	Objective
Lateral	$w_{(lat)} = 2.5$	$\exp\left(-5 \cdot \left(e_{(lat)}/(L_w/2)\right)^2\right)$	Lane keeping accuracy
Heading	$w_{(head)} = 1.0$	$\exp\left(-3 \cdot \left(e_{(\psi)}/(\pi/4)\right)^2\right)$	Heading alignment
Smoothness	$w_{(smooth)} = 0.8$	$\exp\left(-0.5 \cdot \left((\dot{\delta})/0.2\right)^2\right)$	ISO 15622 §9.2 comfort
Progress	$w_{(prog)} = 0.5$	$\left(v_x \cdot \cos\left(e_{(\psi)}\right)\right)/V_{(ref)}$	Prevent zero-speed collapse
Terminal		`$- 10.0$ if	$e_{(lat)}$

The lateral component is dominant ($w_{(lat)}$ = 2.5) because lane keeping is the primary control objective. The smoothness component enforces the comfort requirement of ISO 15622:2018 §9.2 by penalizing large steering rates, with the Gaussian width set to the IEEE 2846-2022 target of $(\dot{\delta})_{(RMS)} < 0.2 rad/s$.

The same reward function is applied to both real-world transitions (dataset adapters) and simulation rollouts, ensuring consistent reward signal across the fused replay buffer.

3.6 Training Protocol

3.6.1 Curriculum Learning Schedule

Training follows a four-phase curriculum that progressively introduces harder road scenarios:

Curriculum Phases Table

Phase	Episodes	Scenarios	Rationale
1	0–150	SCN-01	Straight road baseline
2	150–300	SCN-01, SCN-02	Add constant curvature
3	300–450	SCN-01, SCN-02, SCN-03	Add dynamic curvature
4	450–600	All (SCN-01–05)	Full scenario distribution

Within each phase, scenarios are sampled uniformly at each episode. Initial lateral perturbations are drawn from U(-0.2, 0.2) m for disturbance rejection assessment

3.6.2 Hyperparameter Configuration

Hyperparameters Table

Parameter	Symbol	Value
Actor learning rate	$\eta_{(\mu)}$	$1 \times 10^{(-4)}$
Critic learning rate	η_Q	$1 \times 10^{(-3)}$
Discount factor	γ	0.99
Polyak coefficient	τ	0.005
Batch size	N	256
Policy update frequency	d	2
Critic gradient clip	—	1.0
Warmup steps	—	5000
Total buffer capacity	—	2,000,000
Random seed	—	42

3.6.3 Convergence Monitoring

Convergence is monitored via a 50-episode rolling window of the RMSE lateral error. The agent is considered converged when the rolling RMSE consistently falls below the threshold of 0.40 m for at least 50 consecutive episodes.

Training logs are recorded per episode with 28 fields including: episode return, RMSE $e_{(lat)}$, LKSR, critic/actor losses, Q-value statistics, buffer composition, and noise parameters.

3.7 Evaluation Protocol

3.7.1 Deterministic Policy Assessment

Evaluation employs the trained actor network with noise disabled ($\sigma = 0$). For each of the five scenarios, 20 episodes are executed with random initial lateral perturbations drawn from $U(-0.2,0.2)m$, yielding 100 total evaluation episodes.

3.7.2 ISO 15622:2018 Metrics (M-01 to M-06)

ISO Metrics Table

ID	Metric	Formula	Threshold
M-01	Mean lateral error	$\overline{(e)_{(lat)}} = 1/T \sum (t = 1)^T \left\| e_{(lat,t)} \right\|$	< 0.30 m
M-02	RMSE lateral error	$e_{(lat,RMSE)} = \sqrt{\left(1/T \sum (t = 1)^T e_{(lat,t)}^2\right)}$	< 0.40 m
M-03	Max lateral error	$e_{(lat,max)} = \max_t \left\| e_{(lat,t)} \right\|$	Reported
M-04	Heading RMSE	$e_{(\psi,RMSE)} = \sqrt{\left(1/T \sum (t = 1)^T e_{(\psi,t)}^2\right)}$	< 0.087 rad (5°)
M-05	Lane Keeping Success Rate	$LKSR = \text{(in-lane steps)}/\text{(total steps)}$	≥ 0.95
M-06	Lane Departure Rate	$LDR = \text{(departure episodes)}/\text{(total episodes)}$	< 0.05

3.7.3 IEEE 2846-2022 Metrics (M-07 to M-10)

IEEE Metrics Table

ID	Metric	Formula	Target
M-07	Settling time	Time to $\lvert e_{(lat)} \rvert < 0.10m$ sustained 0.5 s	Reported
M-08	Overshoot	$\max\left(e_{(lat)}\right) / e_{(lat)}(0)$	Reported
M-09	Control effort	$CE = \int_0^T \delta(t)^2 dt$	Reported
M-10	Steering rate RMS	$(\delta)\dot{}_{(RMS)} = \sqrt{\left(1/T \sum(t=1)^T (\dot{\delta})_t^2\right)}$	< 0.20 rad/s

3.7.4 UNECE WP.29 R157 Safety Metrics (M-11 to M-13)

UNECE Metrics Table

ID	Metric	Formula	Threshold
M-11	TTLD series	$TTLD_t = \left(L_w/2 - \lvert e_{(lat,t)} \rvert\right)/v_y$	—
M-12	TTLD 5th percentile	$P_5(TTLD)$	$\geq 0.4\ s$
M-13	Safe Boundary Violation Rate	$SBVR = \left(\text{steps with } \lvert e_{(lat)} \rvert > 0.75m\right) / (\text{total steps})$	< 0.01

3.7.5 Dataset Quality Metrics (M-19 to M-22)

Dataset Quality Metrics Table

ID	Metric	Description
M-19	Real-to-sim ratio	Proportion of real vs. simulated transitions in buffer
M-20	Tyre calibration R^2	Goodness of fit from DS-02 calibration

M-21	Curvature coverage	Distribution overlap of κ across datasets
M-22	Pre-train performance	RMSE $e_{(lat)}$ after pure pre-training (no sim)

3.8 Summary

This chapter has presented a comprehensive methodology for DDPG-based lane keeping that integrates three external driving datasets with a dual-mode simulation environment. The CARLA 0.9.16 backend provides four-wheel PhysX dynamics for high-fidelity evaluation, while the bicycle model enables rapid offline training. The evaluation framework spans three international standards (ISO 15622, IEEE 2846, UNECE R157) with 22 quantitative metrics, ensuring industrial-grade assessment of the autonomous lane keeping controller.